

YASHASWI VEMU

Software Engineer

SUMMARY

AI Developer with experience building generative AI and LLM-powered automation solutions for requirement analysis, workflow automation, and test case generation. Proficient in Python, API integration, and AI-driven system integration, with experience developing prototypes and deploying scalable automation tools in enterprise environments.

PROFESSIONAL EXPERIENCE

AI/ML Engineer, Paves Technologies

Dallas, TX | Sep 2025 – Present

Technologies: Python, PyTorch, LangChain, FAISS, OpenAI API, FastAPI, Docker

- Designed and deployed ML-based transaction ranking and fraud scoring models using Python (pandas, scikit-learn), improving prioritization efficiency across high-volume payment review queues.
- Built supervised learning models leveraging structured transaction data and behavioural features to detect suspicious activity and optimize risk thresholds.
- Engineered feature pipelines and evaluation workflows including cross-validation, precision/recall analysis, and ROC-AUC metrics to improve model performance.
- Conducted controlled experiments to evaluate model configuration changes and optimize fraud detection precision and recall trade-offs.
- Built reusable ML pipeline components for model training, validation, deployment, and monitoring to standardize fraud detection workflows across teams.
- Exposed real-time ML prediction services through Java and Python REST APIs, integrating fraud scores into internal dashboards and operational monitoring systems
- Processed and analysed large-scale verification datasets to support NLP-based document classification and identity verification workflows.
- Designed and maintained scalable backend services on GCP supporting high-volume transaction processing systems.

AI/ML Engineer, ServiceNow

Hyderabad, India | Jul 2022 – Jul 2024

- Designed and maintained scalable backend services supporting enterprise workflow automation using Java and Python.
- Built distributed service components to support upgrade orchestration across 500+ global deployments.
- Developed REST APIs and modular backend libraries enabling automation across platform environments.
- Contributed to automated validation workflows integrated within CI/CD pipelines, ensuring model-driven upgrade decisions met regression and reliability standards.
- Collaborated with cross-functional teams (engineering, QA, product) to deliver scalable automation features supporting enterprise deployments.
- Contributed to automated validation workflows integrated within CI/CD pipelines, ensuring model-driven upgrade decisions met regression and reliability standards.

PROJECTS

Athleta AI – Distributed E-commerce Platform

UTA | Fall 2025

Technologies: Java, Spring Boot, React, AWS SQS, DynamoDB, Docker

- Designed a distributed e-commerce platform using Java Spring Boot microservices and React for product browsing, auth, and order processing built for horizontal scalability and fault isolation.
- Implemented event-driven order workflows with AWS SQS for async order/payment/inventory communication; RESTful APIs backed by DynamoDB and containerized with Docker; achieved 99.9% availability under load.

Visual Question Answering System

UTA | Spring 2025

Technologies: Python, TensorFlow, CNN, LSTM, NLP

- Developed and trained multimodal deep learning models combining CNN and LSTM architectures.
- Performed model training, hyperparameter tuning, and performance optimization.
- Evaluated model accuracy using structured validation pipelines and error analysis.
- Applied supervised learning techniques and experimented with training cycle improvements.

MavPath – Campus navigation app using React and Next.js

UTA | Fall 2025

Technologies: Python, React (TypeScript), Next.js, Node.js, REST APIs, PostgreSQL, Google Maps

JavaScript API, Places API, Directions API, GeoJSON, Jest, Playwright, Vercel, Git, Agile

- Built full-stack web platform using React (TypeScript), Next.js, and REST APIs to support dynamic routing and location-based data queries.
- Designed backend services and PostgreSQL schema to manage structured geospatial data.
- Integrated third-party APIs (Google Maps, Directions API) for real-time route generation.

Large Language Model Training & Optimization

- Experimented with supervised fine-tuning (SFT) workflows to improve task-specific performance of language models.
- Built structured training datasets and preprocessing pipelines for LLM alignment.
- Designed evaluation metrics to benchmark output quality across iterations.

CONTACT

Dallas, TX

(214) 709-0938

vemuyash@gmail.com

[LinkedIn](#)

[Portfolio](#) | [GitHub](#)

EDUCATION

Master of Science(current)

Computer Science

University of Texas at Arlington

Arlington, TX | **Expected:** May 2026

Bachelor of Technology

Computer Science & Engineering

G Narayanamma Institute of Technology

Hyderabad, India | May 2024

Relevant Courses:

Data Structures & Algorithms, Distributed Systems, Machine Learning, OS, Database Systems, Software Engineering

TECHNICAL SKILLS

Languages:

Java(Primary), Python, TypeScript, JavaScript, SQL

Frameworks:

React.js, Spring Boot, Node.js, Express.js, Next.js

Automation & Tools:

Python, Selenium (Exposure), Visual Studio Code, Databricks (Exposure), REST APIs, Test Automation Frameworks

Databases:

Vector Databases (FAISS, ChromaDB), Graph Databases

Cloud:

GCP (Vertex AI, Pub/Sub, BigQuery), AWS (SQS, DynamoDB, EC2, S3, Lambda)

Machine Learning: scikit-learn, XGBoost, LightGBM, pandas, numpy, Feature Engineering, Model Evaluation, Cross-Validation, SHAP, Model Interpretability, Drift Detection

AI & Agents:

Generative AI, Vertex AI, MCP, Fluent SDK, ServiceNow AI Agent Studio, Machine Learning, NLP, Recommendations, Ranking, Fraud Detection, LLMs, Embeddings, Prompt Engineering, MLOps.

DevOps:

Docker, Git, CI/CD, Jenkins, Bash, Linux

Software Engineering Practices: Full SDLC, Production Deployment, CI/CD, Test Automation, Version Control, Code Reviews

LLMs: Prompt Engineering, Embeddings, Context Window Optimization

EVIDENCE OF EXCELLENCE

- Contributed to ServiceNow's enterprise AI Merge Agent reducing manual upgrade conflict resolution using Fluent SDK, MCP, and AI Agent Studio.
- Integrated production GenAI and NLP workflows at Paves Technologies using Vertex AI to automate compliance document processing, reducing manual review overhead at scale.
- Built ML-based fraud scoring and transaction ranking pipelines in Python, surfacing real-time prediction scores through backend APIs in high-volume payment systems.